

# Table of contents

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## Introduction

This document demonstrates automatic cross-references with page numbers. The table of contents above is not filled in by hand. The numbers after each link are rendered in CSS via `target-counter(attr(href), page)`.

The first glu pass does not yet know the page numbers and renders ?. The next pass, after the anchor positions have been written to the aux file, has the real values. glu's multi-pass loop converges automatically.

## Line breaking

The TeX algorithm by Knuth and Plass minimises global badness instead of breaking greedily line by line. boxes and glue ports it to Go.

A paragraph turns into a list of glyph, box, and glue nodes. The solver picks the decomposition with the lowest total cost, which means fewer orphans and widows than greedy CSS line breaking gives you.

## Hyphenation

The hyphenation patterns follow Liang/Knuth. CSS Text 3 `hyphens: auto` and `hyphens: manual` are supported, and soft-hyphens (`&shy;`) are honoured.

## Floats and inserts

Floats can be anchored to the top or bottom of a page via the CSS `float` property. Footnotes use the same insert mechanism but live in their own class with a separator rule.

## Closing remarks

The table of contents at the top of this document is the demonstration of `target-counter()`. Try what happens when you add or remove a section: all following page numbers update automatically as soon as `glue` runs its second pass.